

PROBLEM

Every batch-integration method erases unannotated rare subpopulations, silently.

No existing metric detects this without a prior label.

METHOD: REAL + RareShield

REAL: label-free detectability score $LRS(w) = HM(P_d, P_t, P_n, 1-B)$ over motifs W .

RareShield: motif-preserving integrator; protects LRS without sacrificing major-type alignment.

EVIDENCE

Theorem 1: minimax floor.
Experiment: pancreas, HLCA, macaque retina positive controls.
Scale: Kang 2024 (4.9M cells).
Hierarchical REAL recovers TPEx, ionocyte, LAMP3+ mregDC.

Detect silently. Protect provably. Scale to 4.9M cells.